Well Completion Report — NHK-38

Field: Naharkatia · Block: Upper Assam — Block A
Well type: Development (vertical)
Spud: 14-Feb-2025 · Rig released: 30-Mar-2025
Final depth: 3,712 m MD

1. Summary of drilling

The well was drilled in four phases using 17-1/2", 12-1/4" and 8-1/2" hole sizes. 13-3/8" casing was set at 1,065 m and 9-5/8" casing at 2,448 m, just above the top of the Barail Coal-Shale.

Formation tops encountered: Girujan Clay 1,118 m; Tipam Sandstone 1,941 m; Barail Coal-Shale 2,512 m; Barail Arenaceous 2,851 m; Kopili 3,186 m; Sylhet Limestone 3,431 m.

2. Drilling complications

Partial losses of 45 bbl/hr were observed at 2,395 m while drilling the lower Tipam Sandstone. Pumped 60 bbl LCM pill with 40 ppb calcium carbonate and mica and reduced flow rate. Losses cured after 5 hours and drilling resumed. NPT 6.5 hours.

While making a connection at 2,541 m in the Barail Coal-Shale, the string got differentially stuck. Torque had been rising and erratic for the previous stand, with 35 klbs overpull on the last connection. Spotted 40 bbl pipe-lax pill around the BHA, soaked for 6 hours and jarred down. Pipe freed after 14 hours; resumed drilling. NPT 18 hours.

Heavy splintery cavings were seen on the shakers from 2,590 m in the Barail Coal-Shale. Mud weight was raised from 10.2 to 10.7 ppg and glycol was added for inhibition. Cavings reduced within one circulation.

3. Lessons learned

Pre-treat the active system with LCM before entering the lower Tipam Sandstone. Keep connection times short in the Barail Coal-Shale and watch the torque trend closely; the stuck pipe was preceded by a clear torque increase.